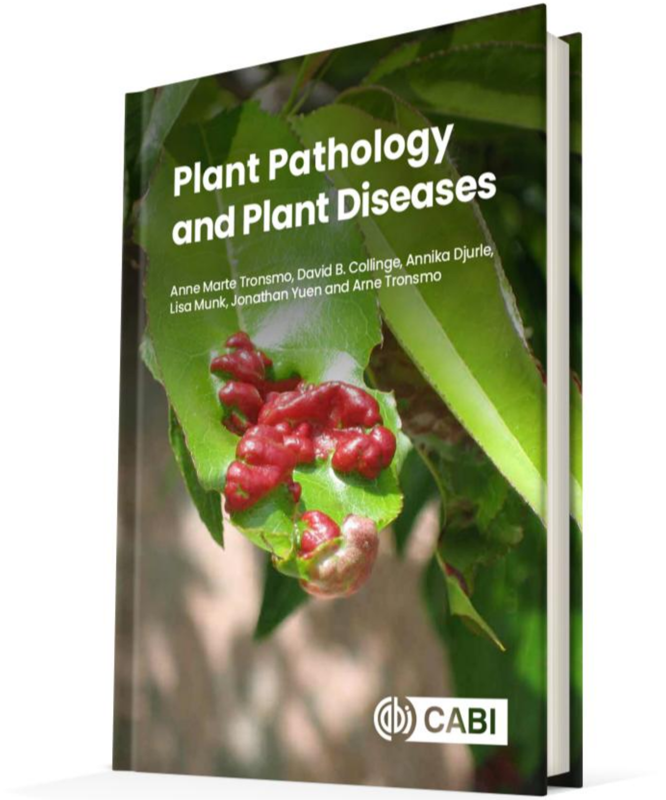

Plant Pathology and Plant Diseases

TEACHING MATERIALS

CABI

Fig. 12.1A-D Resistance genes. (A) and (B) Race specificity in wheat. against *Puccinia striifo*rmis (C) The recessive mutation mlo in barley (*Hordeum vulgare*) confers race non-specific resistance against *Blumeria graminis* f.sp. *hordei* causing powdery mildew. (D) Mlo plants are more susceptible to some hemibiotrophs and necrotrophs such as *Ramularia collo-cygni*. (© D.B. Collinge.)

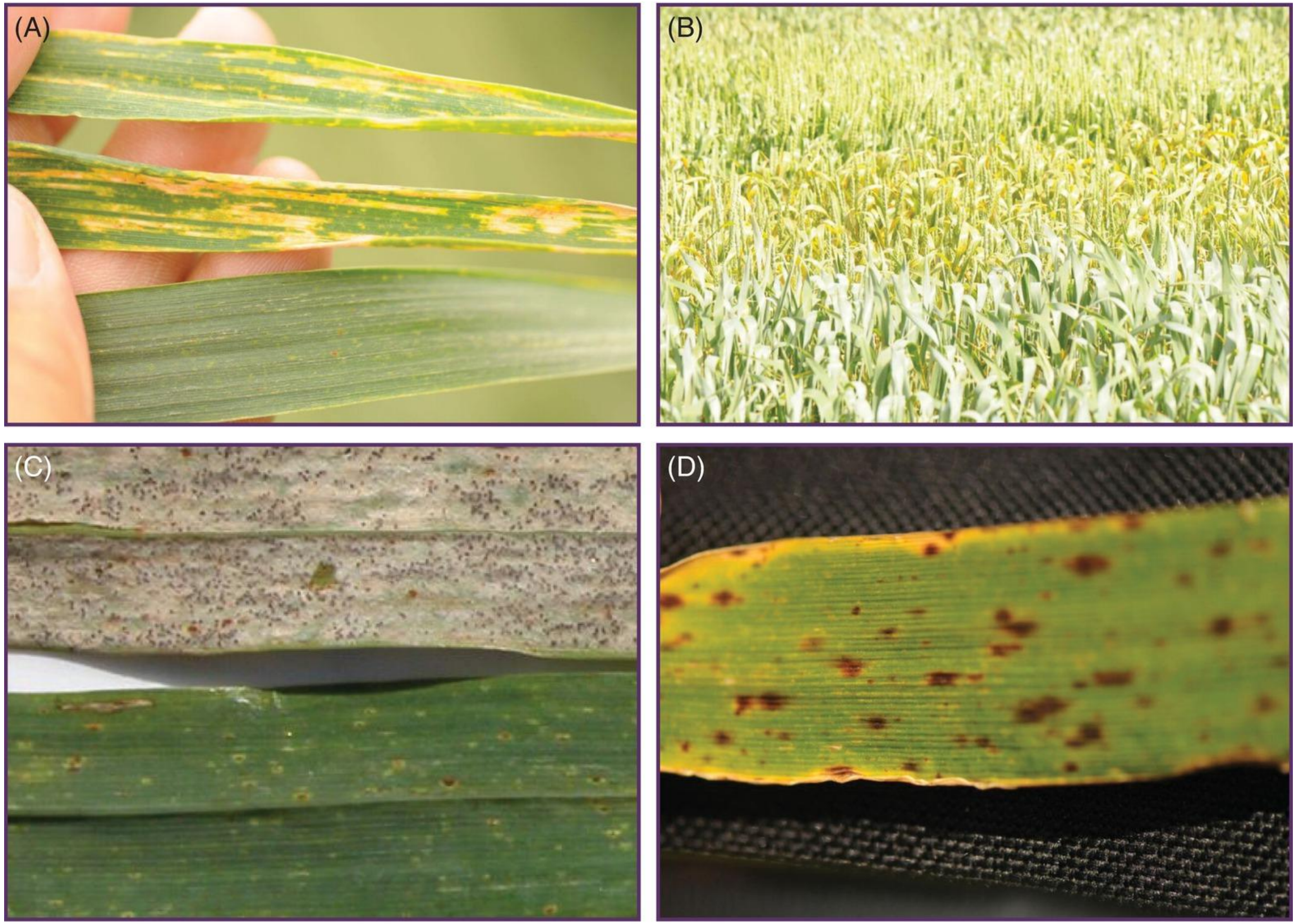

Fig. 12.1 E-F Resistance genes. (E) and (F) Different varieties of wheat exhibit different levels of partial resistance against *Zymoseptoria tritici*. (© D.B. Collinge.)

Fig. 12.2 Layers of plant immunity.
(© D.B.Collinge).

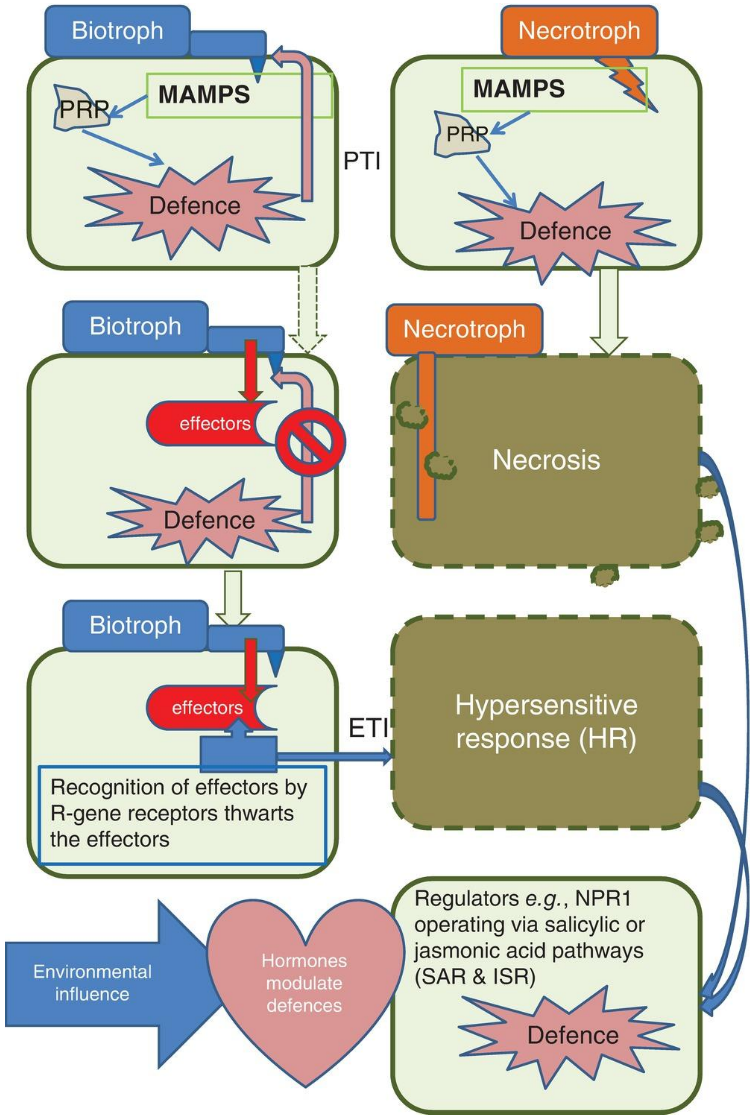

Fig. 12.3 Jones and Dangl (2006) zig-zag model. (D.B. Collinge)

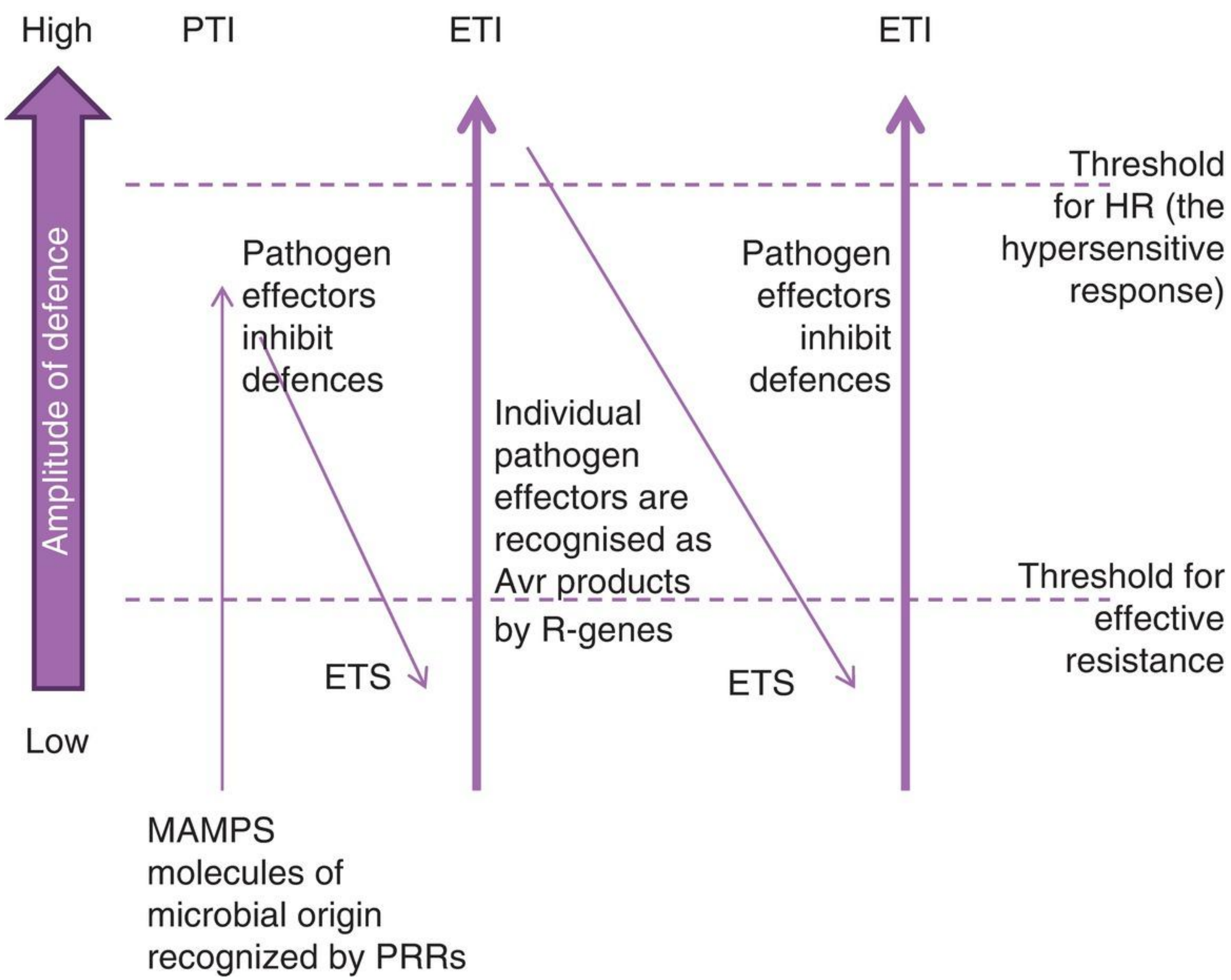

Fig. 12.4 Representative R genes illustrating their modular structures.
(© D.B. Collinge).

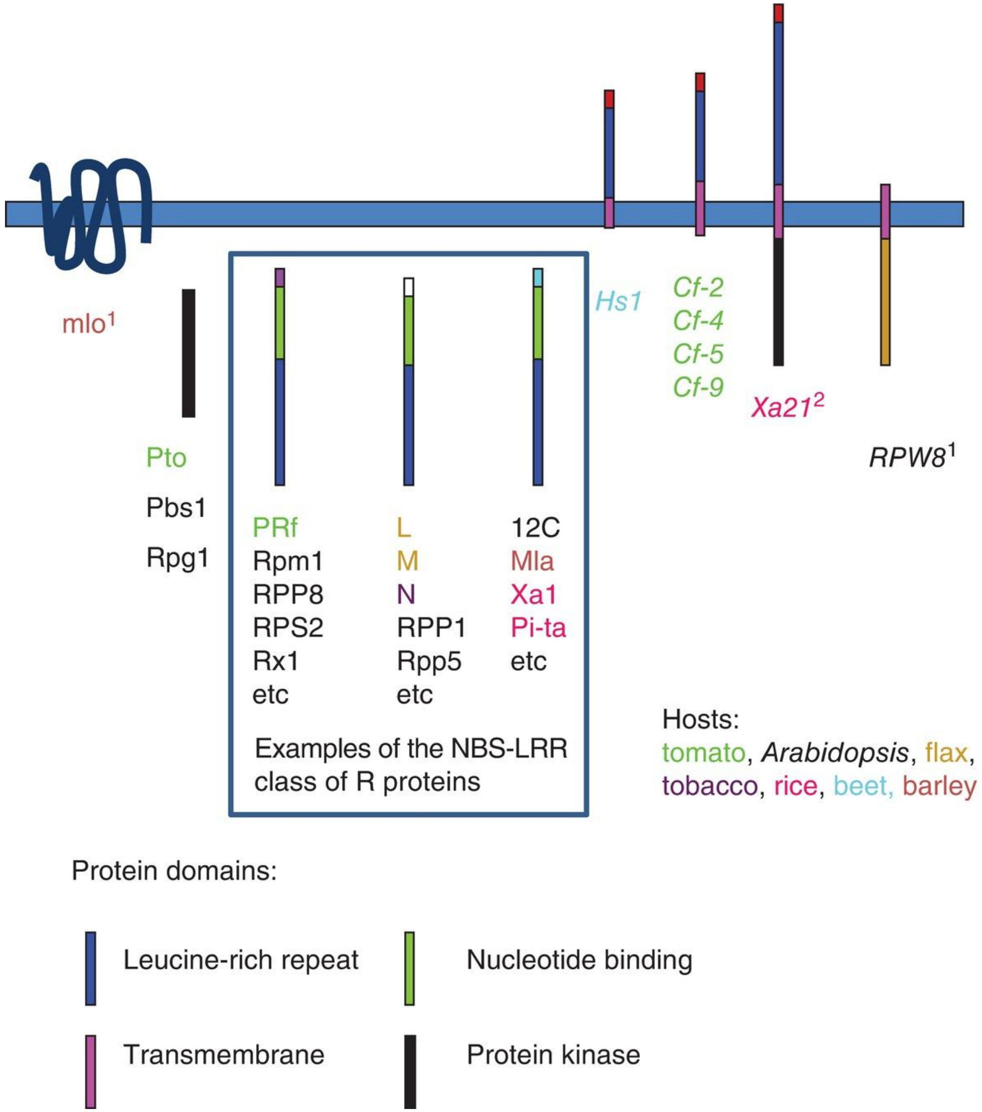

Fig. 12.5 Models for gene-for-gene interactions. (A) Receptor–ligand model. (B) Guard hypothesis. (© D.B. Collinge.)

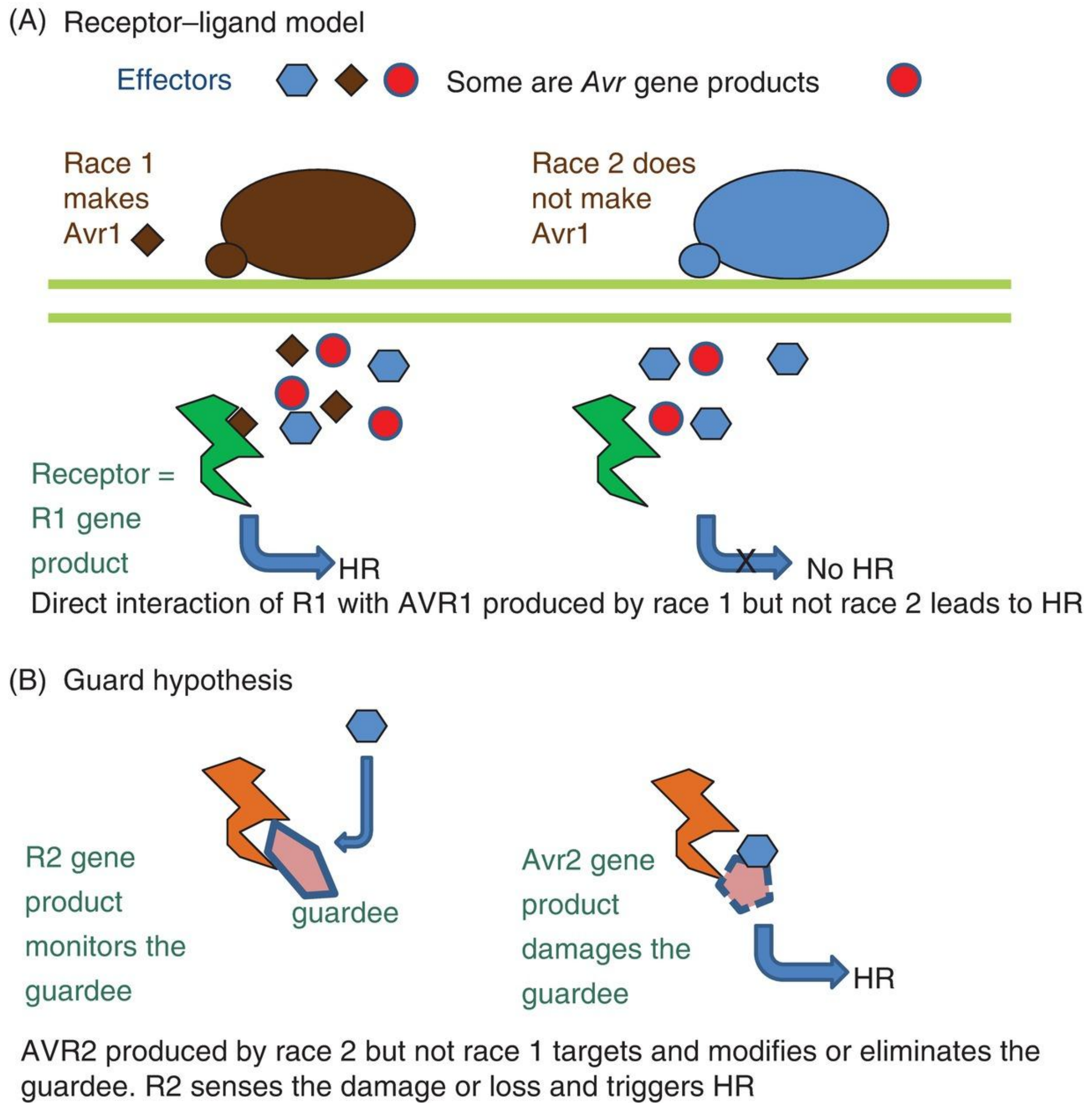

Fig. 12.6 Specificity in the Pto gene of tomato. (A) Pto (a) and Fen (b) genes; (B) Positions of three pto (susceptible) mutations eliminating binding activity in Y2H. (C) Positions of two mutations introduced into Pto eliminating binding activity in Y2H K69N. (A: adapted from Tang et al., 1996; B and C: adapted from Scofield et al., 1996.)